

1. Main entities in a Hospital System

- Patients
- Appointment
- Visit
- Doctors
- Departments
- Medical Record
- Room
- Payment

2. Table Structures (Columns ,Types, Keys)

Patients table

Column	Data type	Key
patient_id	int	Primary Key
name	varchar(150)	
phone	varchar(30)	
email	varchar(150)	

Doctors table

Column	Data type	Key
doctor_id	int	Primary key
doctor_name	varchar(150)	
doctor_email	varchar(150)	
department_id	int	Foreign key- Departments.department_id

Departments table

Column	Data type	Key
department_id	int	Primary key
department_name	varchar(150)	

Appointment table

Column	Data type	Key
appointment_id	int	Primary key
patient_id	int	Foreign key - Patients.patient_id
doctor_id	int	Foreign key - Doctors.doctor_id
status	varchar(50)	
appointment_date	date	

Visit table

Column	Data type	Key
visit_id	int	Primary key
-	int	Foreign key - Patients.patient_id
-	int	Foreign key - Doctors.doctor_id
reason	varchar(150)	
appointment_id		

MedicalRecord table

Column	Data type	Key
record_id	int	Primary key
patient_id	int	Foreign key - Patients.patient_id
past_diseases	varchar	

Room table

Column	Data type	Key
room_id	int	Primary key
bed_number	int	
status	varchar(50)	

Payment table

Column	Data type	Key
payment_id	int	Primary key
-	int	Foreign key - Patients.patient_id
amount	int	
payment_date	date	
appointment_id	int	

3. Primary keys

- Patients.patient_id
- Doctors.doctor_id
- Departments.department_id
- Appointment.appointment_id
- Visit.visit_id
- MedicalRecord.record_id
- Room.room_id
- Payment.payment_id

4. Foreign keys

- Doctors.department_id - Departments.department_id
- Appointment.patient_id - Patients.patient_id
- Appointment.doctor_id - Doctors.doctor_id
- Visit.patient_id - Patients.patient_id
- MedicalRecord.patient_id - Patients.patient_id
- Payment.patient_id - Patients.patient_id

5. Relationship Overview

Patient (1) — (N) Appointment
Doctor (1) — (N) Appointment
Department (1) — (N) Doctor
Appointment(1) — (1) Visit
Patient (1) — (N) MedicalRecord
Patient (1) — (N) Payment
Appointment(1) — (1) Payment

SQL Queries

```
create table patients(  
patient_id int GENERATED ALWAYS AS IDENTITY primary key,  
patient_name varchar(150) not null,  
patient_email varchar(150) not null,  
patient_phone varchar(50));
```

```
create table doctors(  
doctor_id int GENERATED ALWAYS AS IDENTITY primary key,  
doctor_name varchar(150) not null,  
doctor_email varchar(150) not null,  
doctor_phone varchar(50),  
department_id int);
```

```
create table departmentss(  
department_id int GENERATED ALWAYS AS IDENTITY primary key,  
department_name varchar(150) not null);
```

```
create table appointment(  
appointment_id int GENERATED ALWAYS AS IDENTITY primary key,  
patient_id int,  
doctor_id int,  
status varchar(50),  
appointment date);
```

```
create table visit(  
visit_id int GENERATED ALWAYS AS IDENTITY primary key,  
appointment_id int,  
reason varchar(150));
```

```
create table medicalRecord (  
record_id int GENERATED ALWAYS AS IDENTITY primary key,  
patient_id int,  
past_diseases varchar(150) not null);
```

```
create table room(  
room_id int GENERATED ALWAYS AS IDENTITY primary key,  
bed_number int,  
status varchar(150) not null);
```

```
create table payment(  
payment_id int GENERATED ALWAYS AS IDENTITY primary key,  
appointment_id int,  
payment_date date,  
amount number);
```

```
drop table payment;
```

--1

```
alter table doctors
add constraint fk_doctors_departments
foreign key (department_id)
references departmentss(department_id);
```

--2

```
alter table appointment
add constraint fk_appointment_patient
foreign key (patient_id)
references patients(patient_id);
```

--3

```
alter table appointment
add constraint fk_appointment_doctor
foreign key (doctor_id)
references doctors(doctor_id);
```

--4

```
alter table visit
add constraint fk_visit_appointment
foreign key (appointment_id)
references appointment(appointment_id);
```

--5

```
alter table medicalRecord
add constraint fk_medicalrecord_patient
foreign key (patient_id)
references patients(patient_id);
```

--6

```
alter table payment
add constraint fk_payment_appointment
foreign key (appointment_id)
references appointment(appointment_id);
```

```
insert into patients (patient_name, patient_email, patient_phone)
values ('Mochi Cat', 'mochi.Cat@gmail.com', '0501234567');
```

```
insert into doctors (doctor_name, doctor_email, doctor_phone, department_id)
values ('Lala Nuraliyeva', 'lale_N@gmail.com', '0515830651', 1 );
```

```
select * from doctors;
insert into departmentss (department_name)
values ('New Dep');
```

```
insert into appointment (patient_id, doctor_id, status, appointment)
values ( 1, 1, 'ongoing', sysdate);
```

```
insert into visit (appointment_id, reason)
values ( 1, 'chek-up');
```

```
insert into medicalRecord (patient_id, past_diseases)
values ( 1, '-');
```

```
insert into room (bed_number, status)
values ( 10, 'empty');
```

```
insert into payment (appointment_id, payment_date, amount)
values ( 1, '18-MAR-2026', 100);
```

```
insert into payment (appointment_id, payment_date, amount)
values ( 1, '18-MAR-2026', 50.4);
```

--1,2 join

```
select p.patient_name,d.doctor_name,a.appointment,a.status
from appointment a
join patients p
on a.patient_id = p.patient_id
join doctors d
on a.doctor_id = d.doctor_id;
```

```
select p.patient_name,d.doctor_name, d2.department_name,a.appointment,a.status
from appointment a
join patients p
on a.patient_id = p.patient_id
join doctors d
on a.doctor_id = d.doctor_id
join departmentss d2
on d.department_id = d2.department_id;
```

--count

```
select d.doctor_name,count(a.appointment_id)
from doctors d
left join appointment a
on d.doctor_id = a.doctor_id
group by d.doctor_name;
```

```
insert into appointment (patient_id, doctor_id, status, appointment)
values ( 1, 1, 'closed', sysdate-100);
insert into appointment (patient_id, doctor_id, status, appointment)
values ( 1, 1, 'waiting', sysdate+30);
```

--sum

```
select sum(amount)from payment;
```

--cte

```
with avge as(  
select avg(amount)as avg_amount from payment)
```

```
select avg_amount from avge;
```

--view

```
create view patient_appointments as  
select p.patient_name,a.appointment,a.status  
from appointment a  
join patients p  
on a.patient_id = p.patient_id;
```

```
select * from patient_appointments;
```